

Study of the optical properties of solid tissue phantoms using single and double integrating sphere systems

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Abstract Tissue simulators, the so-called tissue phantoms, have been used to mimic human tissue for spectroscopic applications. Phantoms' design depends on patterning the optical properties, namely absorption and scattering coefficients which characterize light propagation mechanisms inside the tissues. In this work, two calibration models based on measurements adopting integrating sphere systems have been used to determine the optical properties of the studied solid phantoms. Integrating sphere measurement results were fed into the calibration models using the multiple polynomial regression method and Newton–Raphson algorithm. The third-order polynomials have been used for optical properties predictions. Good agreement between the two models has been obtained. Role of solid phantoms' components, namely titanium dioxide as a scatterer and black carbon as an absorber, has been discussed. Both of the two components showed observable effects on the absorption and scattering of light inside the solid tissue phantoms.

1 Introduction

Optical properties of the tissues have important role in diagnostic and therapeutic applications of light. Biological tissue could be simulated by adding certain types of particles suspended in a neutral base to mimic events of absorption and scattering. Tissue phantoms may be manufactured in liquid or solid form with suitable contents to give the desired effects.

1.1 Tissue phantoms

Considering the electromagnetic region concerned to study the optical properties, tissue phantoms are widely produced with contents differing according to the required tissue nature. High-water-content tissue could be simulated using gelatin agent with polyethylene powder or water and sodium chloride [1]. Flour, oil, and saline can be used for low-water-content tissues [2]. Also polyester resin, acetylene black, and aluminum powder could be used to mimic bones and human tissues [1–3]. Honey syrup with NaCl simulates human skin [4], while ethanediol, water, salt, and gelatin were used for muscles simulation. For fats ethanediol, gelatin with polyethylene powder [5], or also silicone rubber with carbon fiber can be used [6]. Common choices for scattering particles could be lipid-based emulsion, titanium, and aluminum oxides, in powder form and/or polymer microspheres, while hemoglobin, molecular dyes, ink, and carbon powders are the popular materials used to add the absorption effect into the tissue phantom. Moreover, gelatin, polyester, or epoxy and polyurethane resin were used as base mediums. However, each component has its degree of compatibility with the biological tissues and also some level of stability when reacting with certain light wavelength [7, 8].

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Most of biological and organic materials used in phantoms' formulas are not stable; their chemical composition may change with time. Phantom contents like blood, lipids, milk, oil, or ink have physical and chemical properties that can be stable for hours or few days maximum. Solid phantoms have been studied as good alternatives of short-lived liquid standards [9–11]. The main idea in choosing suitable phantom components is to match the natural biological behavior in reacting with light, which can be tested spectroscopically.

In the present work, solid tissue phantoms have been used. This type of phantoms is interesting for spectroscopy and optical imaging studies through turbid media, improving the determination of optical properties of tissue studies, beam propagation modeling in skin tissue, biomedical optics, and calibration purposes [9–15].

1.2 Integrating sphere

A common simple function of integrating spheres is to measure light attenuation due to directional–hemispherical transmittance, or diffused and luminous reflectance. Theory of light propagation in integrating spheres has been extensively reported [16–18]. An integrating sphere is functioning as a hollow spherical chamber with a highly reflective coating on its internal surface that works to produce homogeneously distributed light inside the sphere. For integrating sphere measurements, samples need minimal amount of preparations. Almost all researches conducted on the integrating sphere theory depend on the measurements of reflectance and transmittance [16, 18–23].

Considering the light beam propagation direction, entrance and exit ports of a single integrating sphere can be used as sample positions for measuring transmittance and reflectance, respectively. In order to take the measurements of transmittance and reflectance simultaneously, a system called double integrating sphere can be used especially when it is important to observe the measurements during a continuous heating, chemical reactions, or applying an electric field. Double integrating sphere setup is supposed to reduce error causes and measurement uncertainties [18]. Theory of double integrating sphere basically depends on the simple theory of single sphere.

Measurements that are taken by the integrating spheres help in understanding the light penetration and light–sample interaction. Moreover, optical properties of solid, liquid, and powder samples can be determined depending on the integrating sphere measurements of reflectance and transmittance. Understanding the light–sample interaction is extremely beneficial to exploit the data given by the integrating sphere and to determine the optical properties.

Since it is not possible to estimate the optical properties of the samples analytically, numerical iterative computer

simulations have been used with integrating sphere measurements. To calculate the so-called scattering coefficient and the anisotropy factor, the value of the collimated (non-diffuse) transmittance is required in addition to reflection and diffuse transmission, while absorption coefficient and reduced scattering coefficient [which is directly proportional to the scattering coefficient with a factor of (1-the anisotropy)] can be determined from measurements of diffuse reflectance, diffuse transmittance, and total transmittance.

Previous literature showed good agreement between the advanced systems analysis program (ASAP) and two single integrating sphere models, one of which was calibrated with He–Ne and the other with white light source at 632.8 nm [24, 25]. In order to prepare the double-sphere system for simultaneous transmittance and reflectance measurements in case of continuous chemical or temperature change, a preliminary test is needed to compare between the double-sphere model and the previously trusted single-sphere model calibrated with white light at 632.8 nm.

The present work depends on previously calibrated systems of single and double integrating spheres [24]. Using solid phantoms, the rapprochement level between values of absorption coefficient determined by double integrating spheres and single-sphere systems has been tested and showed good agreement with the ASAP (Breault Research Organization, Inc.). Similar comparison has been done for the reduced scattering coefficient. Simultaneous monitoring of the change in both of the absorption and reduced scattering coefficients at 632.8 nm for different concentrations of titanium dioxide and black carbon as scatterer and absorber, respectively, allows testing compatibility of the presented type of solid phantoms as standard tissue simulators. Moreover, supportive experiment of imaging the real propagation of the 632.8 nm He–Ne laser beam has been performed to enhance the explanation of the behavior of optical properties as functions of the integrating spheres measurements.

2 Materials and method

2.1 Solid phantoms preparation

In this work, solid phantoms were used as human tissue simulators. The prepared solid phantoms consist of four main components: (1) epoxy resin (Aka-resin, Akasel) with known refractive index equaling 1.55 was used as a base medium, (2) well-defined quantities of Titanium (IV) oxide (TiO₂, anatase, Sigma-Aldrich) were used as scatterer, with a 2.55 refractive index, (3) a black carbon powder pigment has been added in order to act as an absorber of the incident light rays, (4) Aka-cure resin from Akasel was used as

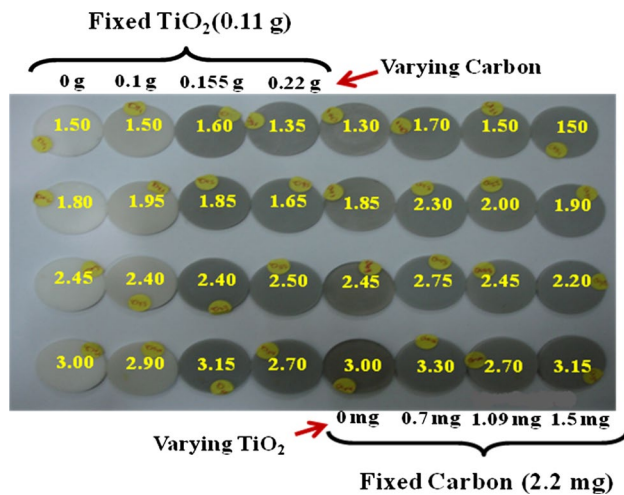


Fig. 1 Stable long-life solid phantom groups: The two groups of phantoms A and B with their carbon and TiO_2 concentrations, also the thickness in mm is written on the phantom disks

solidification material because it becomes solid after being exposed to air.

Two solid phantom groups namely A and B have been prepared. All phantoms in the group A have black carbon powder amount of 2.2 mg, while group B contains phantoms with fixed amount of TiO_2 (0.11 g). Figure 1 shows the phantoms' different thicknesses and the used varying amounts of absorber and scatterer.

The absorbing and scattering powder weights were determined using a digital scale and then added gradually to the base material, and after manual mixing, a

sonicator was used in order to homogenize the liquid mixtures. The high-density liquid has been allowed to dry for approximately 20–26 h in cylindrical holders (Einbettform, Teflon holders), and after drying each cylindrical rigid phantom has been cut into disks of four different thicknesses. Finally, smoothing procedure including cleaning and polishing has been performed. Preparation method of these stable and reproducible phantoms was described in details elsewhere [24, 26].

2.2 Integrating sphere setup

An experimental setup including both single and double integrating spheres has been used, as shown in Fig. 2. A super continuum white light source (SuperK Red, KOHERAS, Denmark) has been used. Light beam was firstly aligned with the 6-inch-diameter spheres (US-060, Labsphere, USA) such that it passes through their centers and the centers of their entrance and exit ports, then aligned similarly with another 8-inch single sphere (US-080, Labsphere, USA) such that the three spheres were coaxial. The signal on each sphere is detected by a 600- μm -core-diameter fiber coupled to the spectrometer (USB4000, Ocean Optics, USA). The detected signals were recorded and processed by a specific software (SpecraSuite, Ocean optics, USA). Reflectance and transmission measurements for each sample were the average of three times measurements, which helps to ensure that the samples have relatively constant uniform transmittance over their surfaces.

Fig. 2 Single and double integrating spheres setup: single and double integrating spheres aligned in a fixed axis and coupled to the same white light source

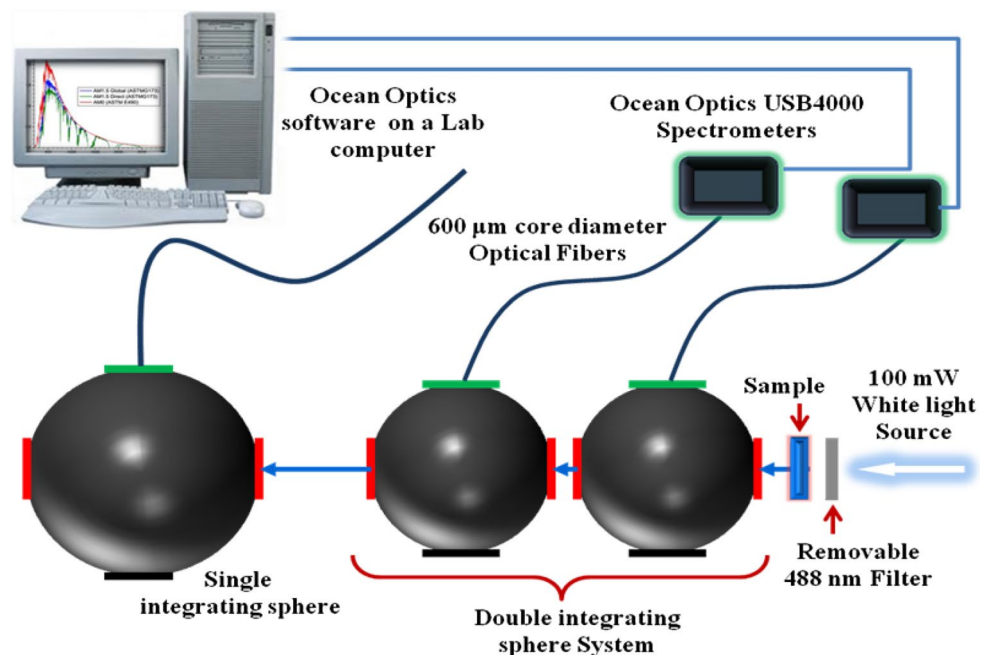
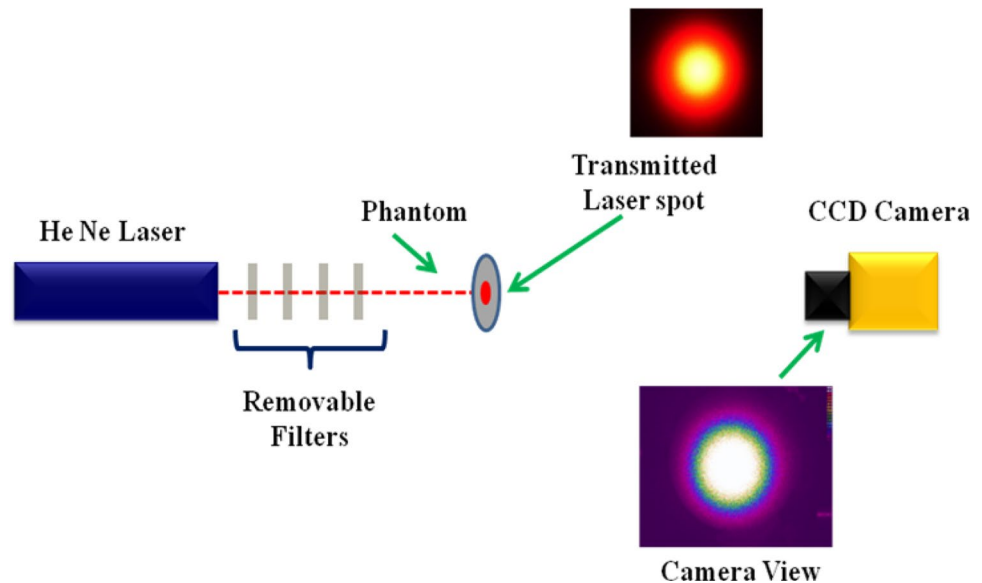


Fig. 3 Imaging setup: Transmitted He–Ne laser intensity from solid phantoms is imaged by a FireWire CCD camera, and suitable power filters are used to produce clear images



2.3 Imaging setup

A 7.15-mW He–Ne laser ($\lambda = 632.8$ nm, JDS Uniphase laser, USA) was used. After passing a set of neutral optical density filters, the laser beam bypasses a pinhole to refine its shape and aligned with the phantom holder in order to strike the phantom centre. Images of the transmitted beam were taken using a CCD camera (Point Grey Scorpion FireWire Camera, Germany) as shown in Fig. 3.

2.4 Reference measurements

Signal should depend only on the sensor area relative to the internal sphere area and opening fraction which contribute directly to the light loss and uncertainties. The additional procedure to provide measurements stability is to normalize the integrating sphere measurements [27]. The detected transmittance and reflectance powers have been normalized for the original power entering the sphere. The reference measurements were taken as shown in the following Fig. 4 using a white cover coated by the same white diffuser used for the internal sphere surface. Reference readings were taken before each separate measurement. On the contrary to [18], the reference measurements of double integrating spheres were taken on the original alignment as previously shown in Fig. 2.

2.5 Calibration models

Both single- and double-sphere systems were calibrated using liquid phantom matrix. The standard phantoms have known and well-calculated optical properties. They were prepared from mixtures of different concentrations of

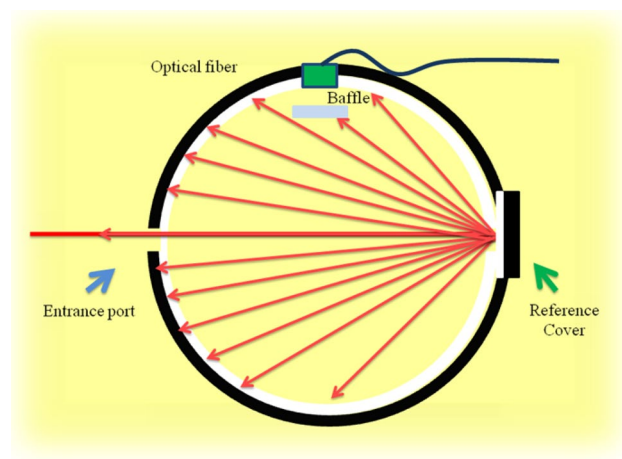


Fig. 4 Integrating sphere reference measurement

intralipid and black ink solved or suspended in fixed water volume.

Optical properties were determined from the measurements using the calibration models, and were extracted using the multiple polynomial regression method [14]. The method involves fitting different order polynomial functions of μ_a and μ_s' to the measured data and to predict μ_a and μ_s' for a particular sample using a Newton–Raphson algorithm. The third-order polynomial has been chosen to fit the measured data to the simulated values to determine the optical properties and the prediction errors. Thickness normalization has been then done for all samples' measurements, by dividing the optical coefficients on the thickness of the original cuvette which has been used for the model construction. Ranges of the calibration models start from 0.55 to 3.28 cm^{-1} for

absorption coefficient values, while the reduced scattering coefficient values were from 0.552 to 110.4 cm^{-1} .

3 Results and discussion

Diffuse reflectance and both diffuse and total transmittance have been measured for all phantoms. Each experimental value is the average of three measurements at three different positions on the phantom disk to confirm its homogeneity. Calculated errors for the integrating spheres' measurements were always less than 0.55 %, which insures the homogeneity of the tested phantoms.

3.1 Absorption coefficient

Figure 5 shows the change in the absorption coefficient with increasing the scatterer amount for phantoms containing the same black carbon powder concentration (2.2 mg). The four phantoms are presented in four subfigures relative to their TiO_2 white powder amounts (a) 0.00 g, (b) 0.10 g, (c) 0.155 g, and (d) 0.22 g. Curves show good stability in the absorption coefficient values of each phantom with thickness change. The absorption coefficient became higher with the increase in white TiO_2 powder in the phantom. Existence of the scattering particles is relevant to multiple deflection of light from its own path. Therefore, the

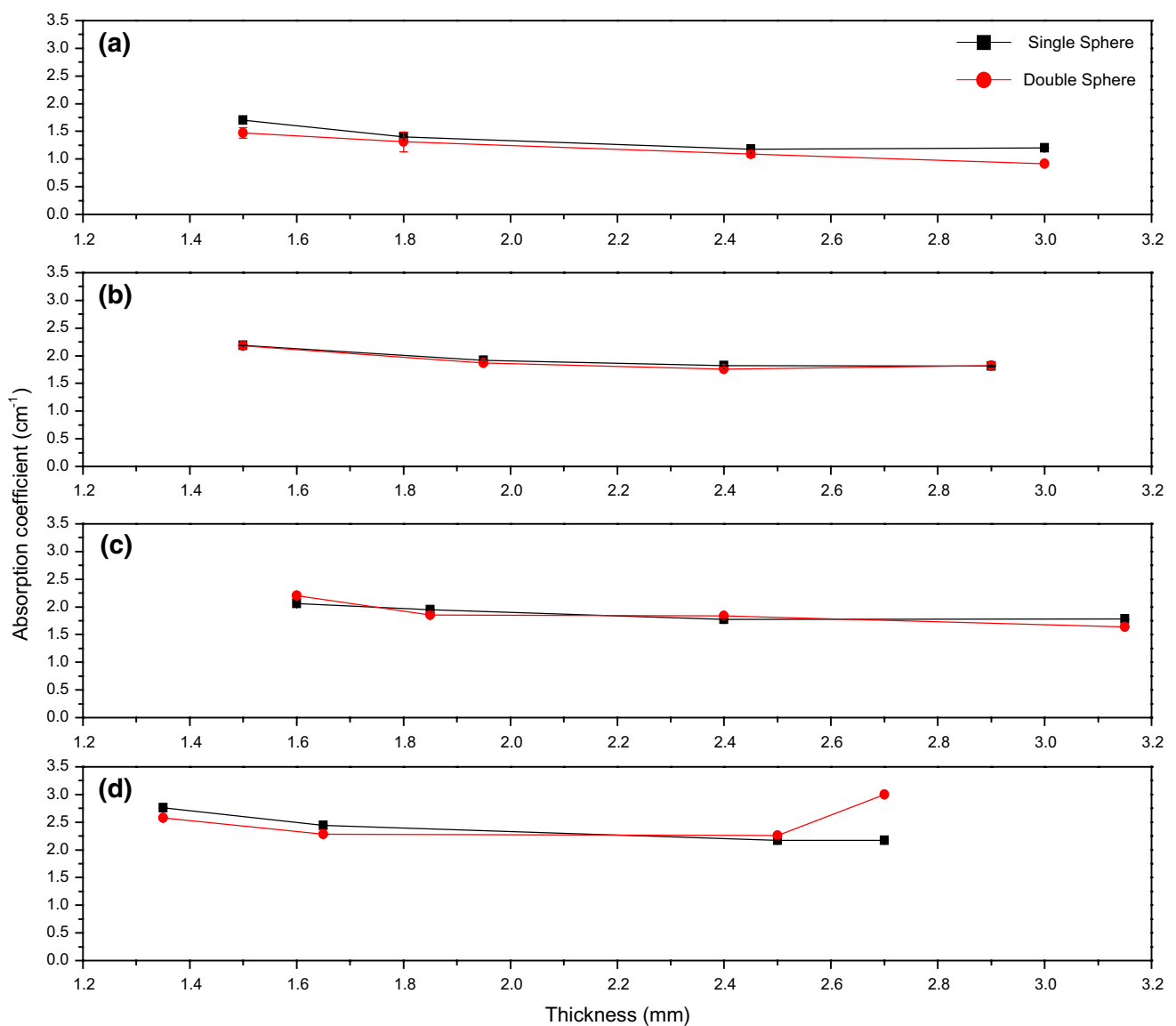


Fig. 5 Absorption coefficients for phantoms with different thicknesses and TiO_2 content **a** 0.0 g, **b** 0.1 g, **c** 0.155 g and **d** 0.22 g, with fixed Carbon amount (2.2 mg)

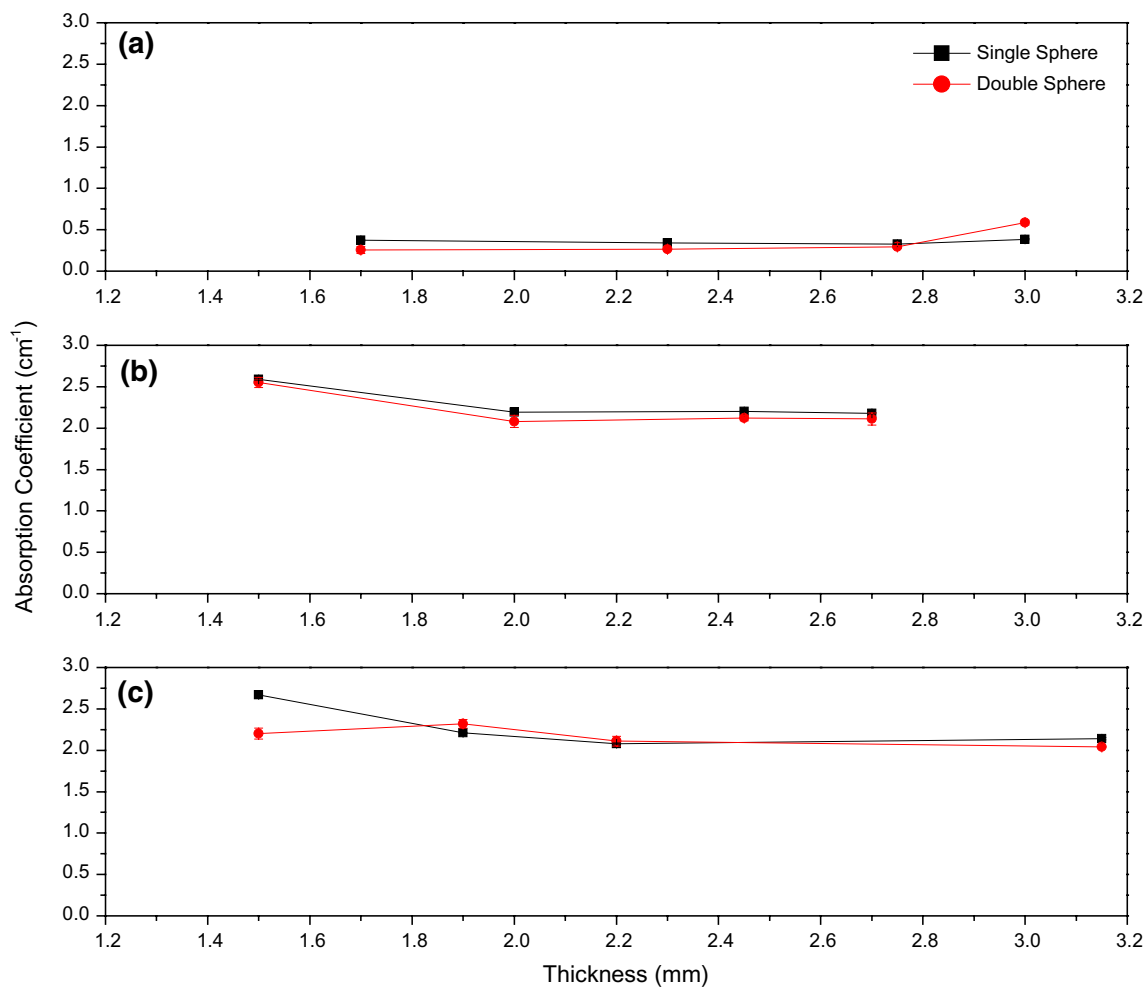


Fig. 6 Absorption coefficients for phantoms with different thicknesses and black carbon content, **a** 0.70 mg, **b** 1.09 mg and **c** 1.50 mg, with fixed TiO_2 amount (0.11 g)

probability of meeting absorbing particles becomes higher. Also, TiO_2 may have some absorptivity at this wavelength.

Absorption coefficient curves of the second group B which contain the same TiO_2 amount (0.11 g) are shown in Fig. 6. Three phantoms are presented with different black carbon amounts: (a) 0.70 mg, (b) 1.09 mg, and (c) 1.50 mg. Determination of the absorption coefficient in case of no carbon black powder added (no absorber) has failed, may be because its value is out of the calibration models. It was observed that as the concentration of the black carbon increases, the absorption coefficient reasonably increases, whereas absorption coefficient showed high stability with thickness change for solid phantom that contains high carbon amount. It can be said also that for the studied absorption coefficient range, there is an observable correlation and compatibility between the two calibration models.

3.2 Reduced scattering coefficient

Figure 7 depicts the reduced scattering coefficient versus the samples' thicknesses for the group A which contains fixed black carbon amount (2.2 mg). Increasing the scatterer amount from 0.00 to 0.22 g induced reasonable increase in the reduced scattering coefficient which is directly proportional to the scattering coefficient. Figure 8 shows the relation between black carbon amounts that change from 0.0 to 1.5 mg, in the B phantom group and the reduced scattering coefficient. Calibration models that depend on both single- and double-spheres systems coupled to the white light source shows direct proportionality between the amount of carbon and scattering coefficient. This was not expected and could be due to aggregation of the carbon particles to a size comparable to 632.8-nm wavelength, such that aggregated particles scatter the

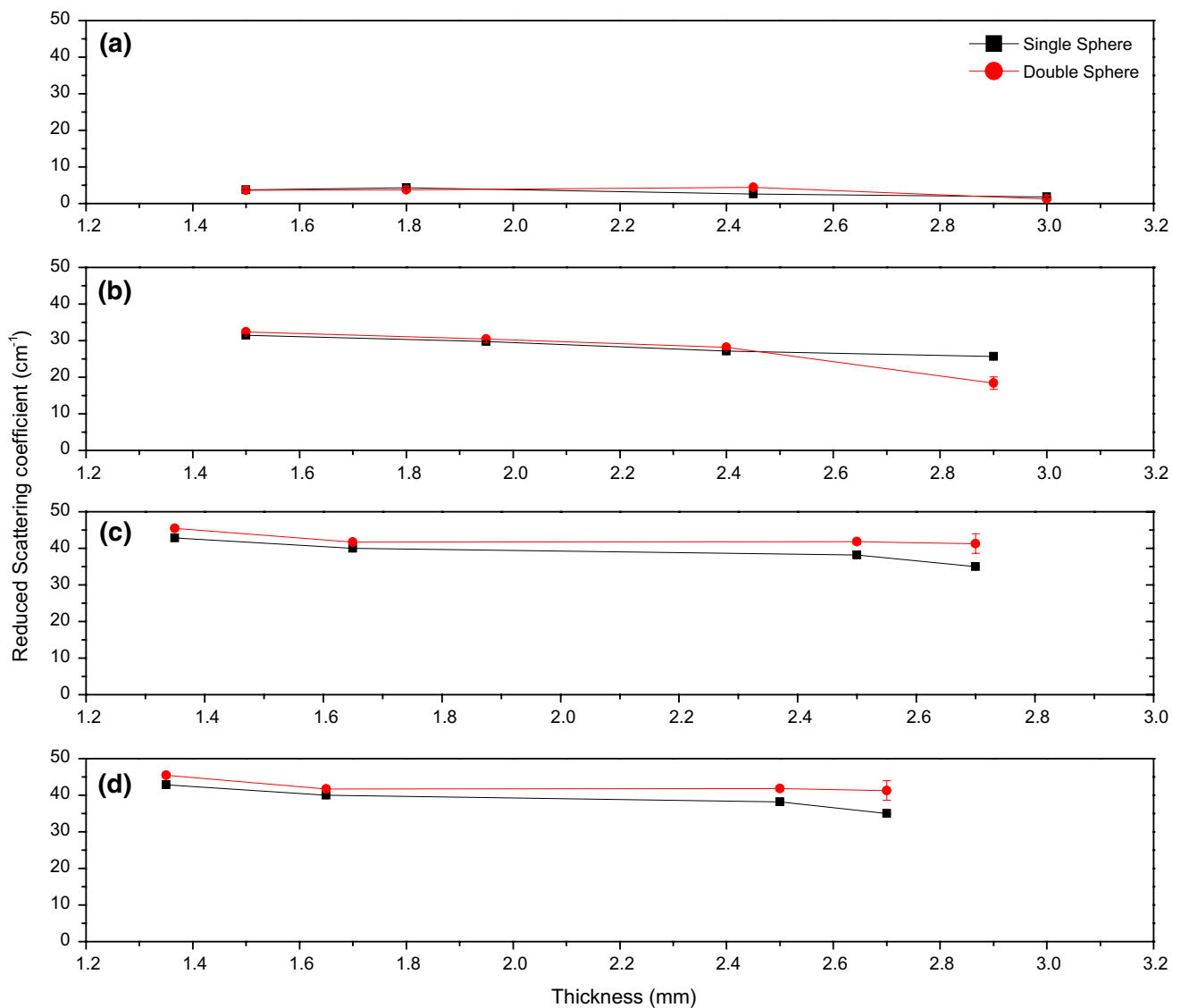


Fig. 7 Reduced scattering coefficient versus the samples' thicknesses for the fixed carbon phantoms (2.2 mg) with TiO_2 **a** 0.0 g, **b** 0.1 g, **c** 0.155 g and **d** 0.22 g

incident photons. In addition, error in the models for the prediction of the scattering coefficient is greater than that for absorption coefficient [24].

3.3 Imaging light penetration

Simple imaging experiments were performed to study light propagation through the solid phantoms of the same laser wavelength at which the optical properties of the samples have been determined in order to enhance the results obtained by the integrating spheres. A 0.82-mm-diameter He–Ne laser beam has been used for imaging the magnification of the beam passing through the examined phantoms. All solid phantom disks have an equally fixed diameter of 30 mm. Using measurements of the imaged

diameters of both solid phantom and the beam passing through it, a simple ratio calculation can lead to the exact passing beam diameter for each case. Then beam magnification is clearly the diameters ratio of the passing beam (which is assumed to maintain the original Gaussian beam profile) to the original He–Ne laser beam. As shown in the figures of both absorption and reduced scattering coefficients (Figs. 4, 5, 6, 7, 8), optical properties showed good stability with thickness change. Unlike the optical properties, beam magnification depends on phantom size and thickness; it is the overall result of absorbing and scattering particles concentrations effect in addition to beam path length. Figure 9 demonstrates the relation between the three parameters (absorber, scatterer, thickness) and the passing beam magnification. Phantom thickness is written

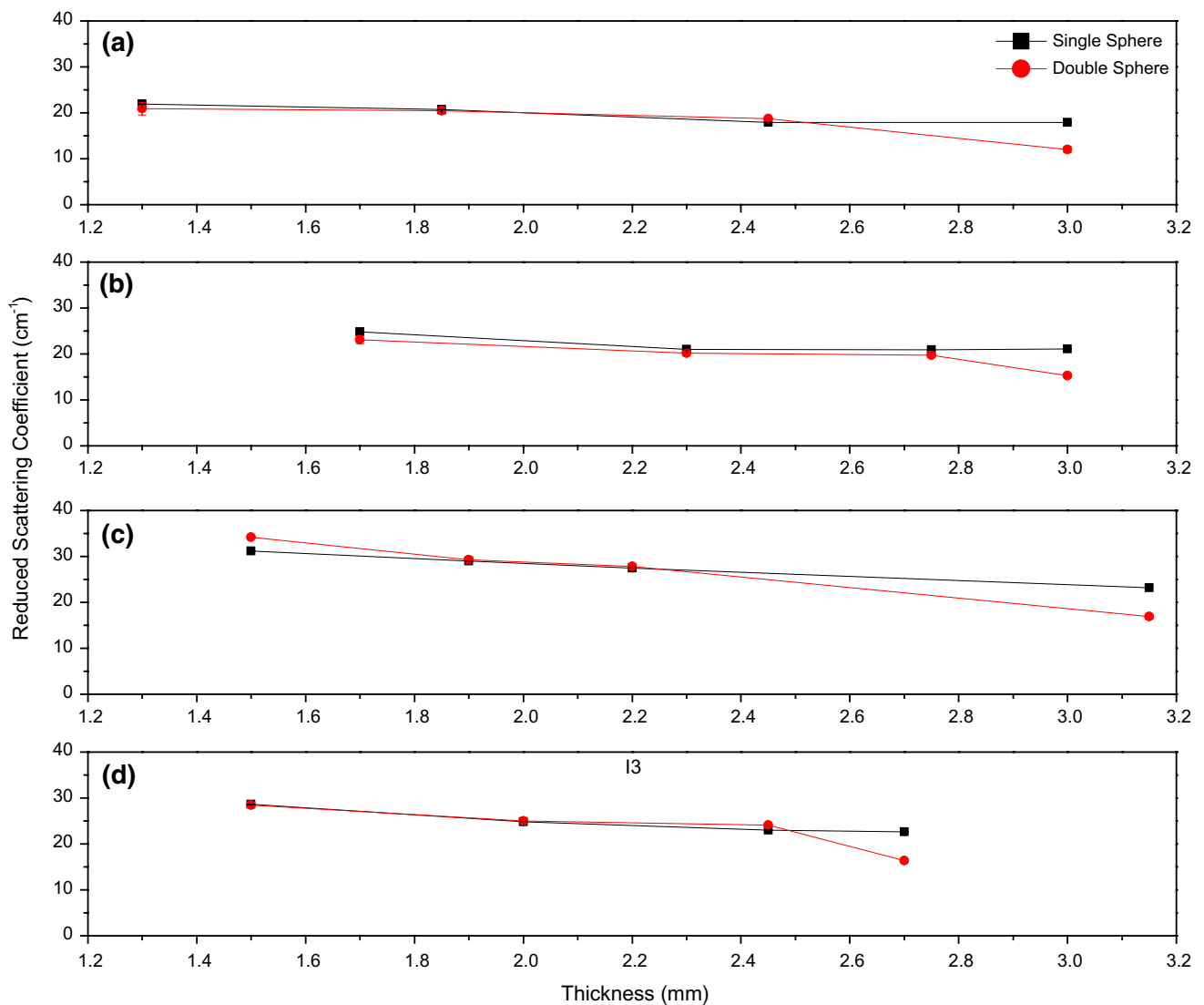


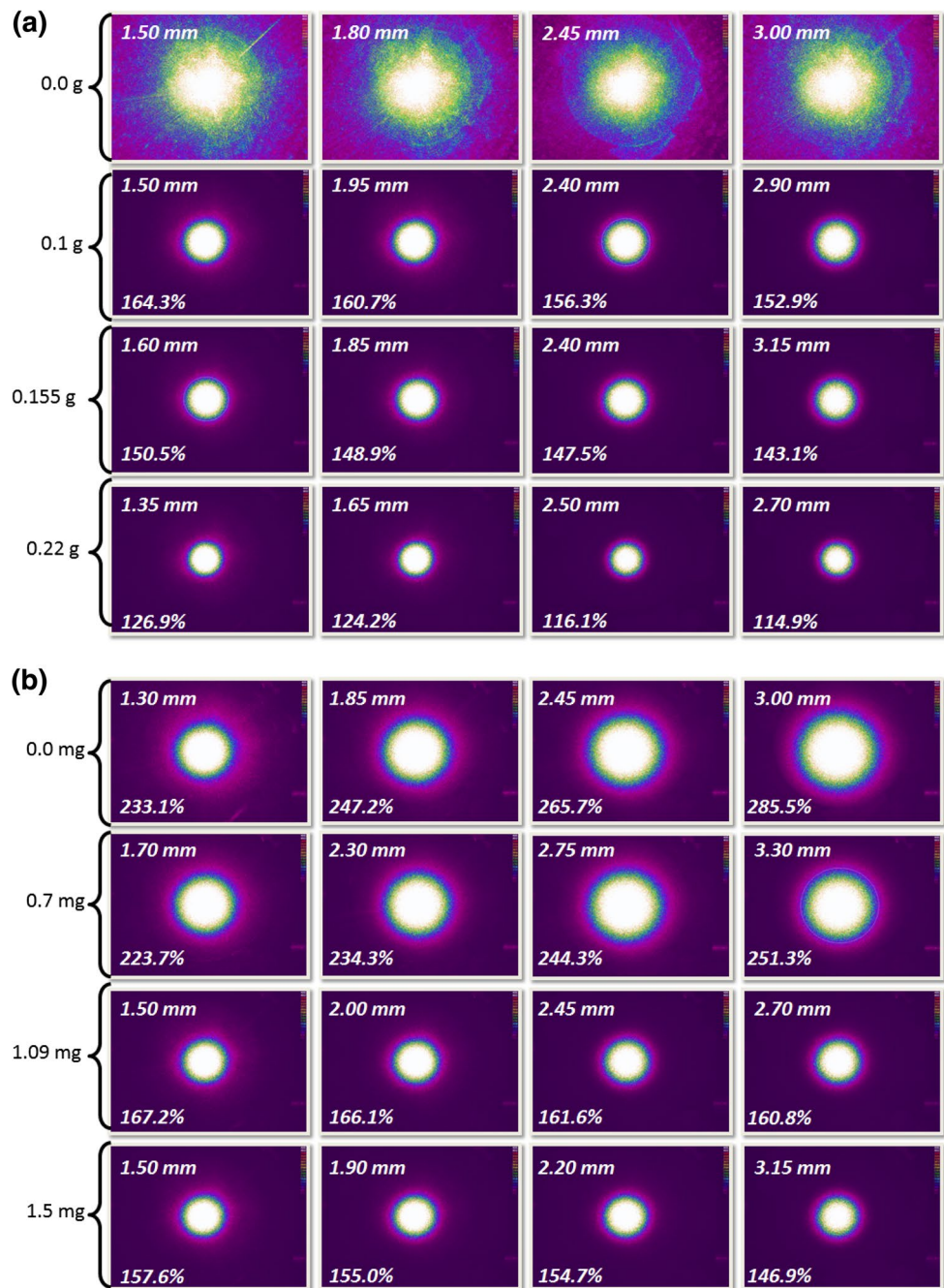
Fig. 8 Reduced scattering coefficient versus the samples' thicknesses, the fixed TiO_2 amount (0.11 g), with black carbon **a** 0.0 mg, **b** 0.7 mg, **c** 1.09 mg and **d** 1.5 mg

above each image, while the magnification percentage is given below. For group (A) phantoms, the passing beam became smaller in diameter with increasing the scatterer amount (downward). That is to say, for the dark A phantoms, addition of the scatterer particles spreads the beam and raises the probability of the photon to strike an absorbing particle; this clearly agrees with the increase in absorption coefficient with adding more scatterer amount which is previously shown in Fig. 5, while in the same time the reduced scattering coefficient increases due to increasing the number of scattering particles as shown in Fig. 7, and therefore the overall result is a slight decrease in the beam magnification because of the competition between absorption and scattering processes. It can be said that these phantoms are more absorptive. Therefore, excess

of absorber in dark phantoms resists beam spreading and force beam magnification. Moreover, beam magnification decays with thickness according to the effect of absorbing particles along the beam path inside the phantom. Set of power filters have been used, in order to get a clear detectable image for the passing beam. Addition of more filters leads to blur almost of the beam images. On the other hand, beam images of the semitransparent phantoms (containing no TiO_2) in the A group appear saturated because of its very high transmittance as shown in Fig. 9a.

Phantoms group B can be divided into two sets: The first includes phantoms with carbon amounts 0.0 and 0.7 mg which may be called the bright or light phantoms. In this set, scattering effect of white TiO_2 particles is stronger than the absorbing particles' effect. Therefore, light spreads

Fig. 9 Passing beam magnification from the solid phantoms groups: imaging the laser diameter passing from solid phantoms with fixed amounts of **a** black carbon (2.2 mg) and **b** TiO_2 (0.11 g). Calculated magnification (on the *bottom*) and phantom thickness (on the *top*) are written



inside and the passing magnified beam increases in size with increasing thickness, while the net result of absorption causes decay in the propagating beam magnification. Similarly as given by the integrating spheres calibrations (Fig. 6), it can be observed that in the first two sets, the scattering is the effective parameter. In the second set, there are phantoms with 1.09 and 1.5 mg carbon concentrations and can be called dark phantoms. This set behaves like the first group of phantoms A where the effect of the absorbing particles is stronger. In addition, light propagation has higher chance to be absorbed along the phantom than to be

scattered back or transmitted. It is also clear that the diameter of the magnified beam decreases (downward) with increasing amount of black carbon in the whole phantom group B as shown in Fig. 9b.

4 Conclusion

In conclusion, single and double integrating sphere systems have been used successfully for transmittance and reflectance measurements of the prepared solid tissue

phantoms. Measurements have been fed into the different calibration models. Optical properties at the 632.8-nm wavelength have been obtained using the multiple polynomial regression method and the Newton–Raphson algorithm. The three different factors that have been studied for the solid phantoms as tissue simulators were: titanium dioxide effect as a scatterer, black carbon as an absorber, and the phantom thickness. Effect of each factor on the light propagation inside the phantoms has been demonstrated and discussed.

Calibration systems showed good agreement with each other, especially for absorption coefficient. Absorption coefficient as well as the reduced scattering coefficient showed good stability with thickness change for all phantoms. It has been found that titanium dioxide has clear effect on light propagation inside the phantoms as appeared in magnified beam images. The spread of the beam inside the phantom raises the probability for a photon to meet an absorbing particle, and therefore absorption coefficient increases. The presence of titanium dioxide leads to high backward scattering. For tissues simulation, backward and forward scatterings must be simulated. Therefore, it may be better for the tissue simulator phantoms to contain many scattering particles types beside titanium dioxide to simulate forward scattering.

Black carbon powder has a reasonable effect on decreasing the transmittance, and therefore an obvious increase in the absorption coefficient has been shown. But it has been observed that even reflectance and reduced scattering coefficient increase with more carbon. This may be because of the possible aggregation of the carbon particles to a size comparable to the used 632.8-nm wavelength.

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